One Operator, Measured Exactly: Cost, Learnability, and the Guidance Frontier in a Single-Operator Representation of Mathematics

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Abstract

One binary operator can, in principle, represent all of elementary mathematics: Odrzywołek (2026) shows every elementary function is constructible from the constant 1 and $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$. The resulting single-operator form is maximally regular, which makes it a natural candidate for graph learning, and we deliver its first exact cost profile at production scale: a fixed corpus of 250,000 deterministic, quality-audited expressions with frozen splits, compiled with the pinned official compiler. The median tree expansion factor is $\alpha = 40.66$ over the source AST, above the preregistered per-family counting thresholds (1.29–1.50) in every family. Lossless DAG sharing recovers a 39.4× mean compression of pure-EML trees (the shared DAG averages 8.3× the uncompressed AST tree), and bounded, verified equality saturation improves a further 23.9–27.6% of costed expressions at 2.4–2.8% mean savings on 60.7% cost coverage; in two preregistered comparisons, equal-budget transparent baselines remain unbeaten by learned components. On this foundation we release a frozen, leakage-controlled learning apparatus and report its first verdicts. The representation is learnable despite the 40× expansion: on binary equivalence, two of three seeds beat the label-imbalance chance floor of ≈0.58 by 0.04–0.05 nats. The preregistered diagnostics earn their place as well: on a goal-conditioned value head whose regression error looks unremarkable, the rank and out-of-distribution probes reveal chance-level ranking (test-IID Spearman ≈0) and a constant out-of-distribution predictor, a guidance signal that error metrics alone would have certified. The cost profile, apparatus, and verdicts are released for reuse.

1 Introduction

Is one operator enough to represent mathematics for a learning model? Every machine encoding of mathematical language must decide how much symbol vocabulary to spend for how much structure; the same trade governs formula tokenization for sequence models and operator-tree granularity in math-aware retrieval (Lample and Charton, 2020; Mansouri et al., 2019). The single-operator representation is the limiting case of that trade, and it is maximally regular: a graph network reading a pure-EML term never has to learn *what* an operator is, only *where* things connect. The objection is size. Lowering even $\sin(x)$ to pure EML produces hundreds of nodes, and the standard reply is that sharing (DAGs), macros, and semantic rewriting could close the gap. Whether they do is an empirical question that, to our knowledge, has not been measured exactly at nontrivial scale. Whether any remaining gap buys something as inductive bias is a second question the measurement alone cannot settle. We answer the first question exactly and report first learning verdicts on the second. The cost is now precisely known; the representation is learnable despite it; and the preregistered diagnostics show what remains to be won, namely usable guidance.

This paper is deliberately narrow. It is a structural feasibility study under preregistered thresholds and complete denominators, and it reports every preregistered outcome exactly. It separates what the representation costs structurally, which we measure in full at production scale, from whether that cost buys anything for learning, which we answer only in part. We fit no scaling law, run no leaderboard, and use no frontier LLM as a controlled baseline; every number below matches a committed per-goal summary exactly.

Contributions.

- The first exact structural cost profile of the single-operator representation at production (250,000-expression) scale: median expansion $\alpha = 40.66$, the full distribution, exact per-family comparisons against preregistered counting thresholds, and content-hashed controlled comparisons in which equal-budget transparent baselines remain unbeaten by learned components (§6).
- The first learning verdicts for the representation: learnability on binary equivalence, with two of three seeds clearing the imbalance-adjusted ≈0.58 chance floor by 0.04–0.05 nats despite the 40× expansion (§6.5).
- A transferable methodological result: preregistered rank and out-of-distribution diagnostics that catch a guidance signal error metrics alone would certify. This is the classical learning-to-rank distinction (Burges et al., 2005) made operational inside a verifier-gated math-search apparatus, which we release for reuse (§§4–5).

2 Related work

Encoding mathematics for learning. Mathematical language admits many machine encodings: token sequences over prefix or infix serializations (Lample and Charton, 2020; Saxton et al., 2019), operator-tree and layout embeddings for formula retrieval (Mansouri et al., 2019), and term graphs with explicit sharing, over which Allamanis et al. (2017) learned equivalence representations on small hand-designed boolean and polynomial corpora; our equivalence task poses the same relation over a 250,000-expression corpus under verifier-gated labels. Every such choice trades symbol-vocabulary size against form length, as subword tokenization does for text. The single-operator representation (Odrzywołek, 2026) is the extreme of that trade, vocabulary collapsed to one operation, and to our knowledge it had no measured cost profile before this study.

Sharing, rewriting, and compression. The recovery mechanisms we test are classical: hash-consed structural sharing (Ershov, 1958; Filliâtre and Conchon, 2006) over standard term graphs (Barendregt et al., 1987), equality saturation for semantic rewriting (Tate et al., 2009; Willsey et al., 2021), and MDL-scored vocabulary induction (Rissanen, 1978). H2 and H3 (§3) instantiate each as a preregistered recovery hypothesis and measure it exactly rather than assuming its benefit.

Learned guidance for symbolic search. Neural premise selection (Irving et al., 2016), language-model proof search with learned value functions (Polu and Sutskever, 2020; Lample et al., 2022), guided synthetic geometry (Trinh et al., 2024), and symbolic regression (Udrescu and Tegmark, 2020) all consume a guidance signal routinely validated by proxy regression losses. Ranking research has long separated regression error from ranking quality (Burges et al., 2005); our value-head diagnosis (§6.5) is a controlled instance of that gap inside a math-search apparatus.

Relation to LLM mathematics. Benchmark-driven studies of large models' mathematical competence (Hendrycks et al., 2021) and natural-language theorem proving (Welleck et al., 2021) evaluate end-task reasoning of frontier systems; we work a layer below, on the representation itself, with compact fixed-budget models, and no LLM enters any controlled comparison.

3 Representations, immutable data, and hypotheses

3.1 Representations

Every representation we compare derives from the same source expression, so differences between them are differences of encoding rather than of content. The structural study (§6) uses four: the ordered source AST as a tree and as a hash-consed DAG (Ershov, 1958; Filliâtre and Conchon, 2006), and the official-v4 pure-EML term (produced by the pinned version-4 reference compiler) as a tree and as a hash-consed DAG. Writing α for the ratio of exact pure-EML tree nodes to source AST nodes, the tree pair fixes α and the DAG pair determines what maximal sharing recovers from it.

The learning grid uses a different, smaller set of representations, because two of the structural objects are too large to encode at production scale. Four graph channels enter the six-arm equivalence grid (§5): the AST DAG, the pure-EML DAG, a macro-derived frequent-motif DAG, and a train-only motif-AST fair control. Two non-graph controls complete the grid: a prefix token sequence over the same expressions, and a floor of 19 predeclared surface counts behind a single linear layer. The floor measures how much of the task surface counts alone already solve; its feature list is frozen in code and asserted by tests.

Macro and motif graphs are lossless DAGs over source constructions admitted to the frozen vocabularies, not pure-EML graphs: each macro node carries `is_pure_eml = false`, and its pure-EML expansion cost is separate metadata never reported as macro size or α. All graphs follow the standard term-graph model (Barendregt et al., 1987), with retained repeated references (so $f(x, x)$ carries two child references to one interned node) and a canonical node signature; structural identity is hash equality on that signature. Semantic equivalence is a separate relation, established only by the verifier, and we mark each use.

3.2 Fixed data foundation

The v1 corpus contains 250,000 expressions, unique under the canonical `(domain_mode, sympy_srepr)` identity (Meurer et al., 2017) rather than up to semantic equivalence, with frozen splits of 175,000 / 25,000 / 25,000 / 25,000 (train / validation / test_iid / test_ood) and six source families under exact quotas: algebraic core (70,000), exp/log (40,000), powers/division/rationals (40,000), trig/hyperbolic (40,000), mixed elementary (35,000), and an OOD-stress family (25,000).[1] Generation is deterministic and QA-gated; manifests carry per-shard checksums, config hash, seed, and commit.

The held-out split and the OOD family coincide by construction: all 25,000 OOD-stress rows are assigned to `test_ood`. Their profile is a combined stress over size, depth, variable count, and composition—not a depth-controlled holdout, and we do not report it as one. This study is fixed to this immutable v1 corpus.

[1]Supplementary bundle: corpus QA report and Goal 2 summary, "Corpus and denominators".

3.3 Predeclared outcome families and hypotheses

Outcome families were declared before the production runs and reported separately—*structural* (tree/DAG size and depth, pure-EML α, macro size, dictionary-inclusive motif description length (Rissanen, 1978)), *predictive* (the six-arm equivalence grid), *rewrite/proof*, *SR*, and *grammar-v2 conformance*. No pooled score exists across families or across channels within the structural family: every structural metric is named per channel, and the plotting code refuses to emit a shared α column.

Three preregistered hypotheses motivate the structural program. **H1**: raw pure-EML expansion stays below the per-family counting thresholds $\alpha_{\text{tree}} < 1 + \ln K/\ln(4L)$ (bounded vocabulary K, literal bound L; range 1.2914–1.5000). **H2**: exact structural sharing closes whatever gap H1 leaves, so a shared pure-EML DAG reaches parity with the uncompressed AST tree. **H3**: verified semantic rewriting and learned motif vocabularies close the residual. Structural statistics use all 250,000 rows; semantic auditing draws on a separate bounded sample and we never extrapolate it (§8).

4 Compact models and fair compute

This section and §5 describe committed, CI-tested apparatus; no number in them is a result (the learning verdicts are in §6.5). We describe the apparatus because its frozen design, not its output, is a contribution.

4.1 A compact GINE-style graph encoder

The encoder is a compact variant of GINE (Xu et al., 2019; Hu et al., 2020): three layers, hidden width 64, and about 0.2M parameters. It uses residual connections, GraphNorm or LayerNorm (fixed per configuration), dropout, a two-layer FFN, and sum pooling. For a message edge $u \to v$ the edge feature and update are

$$a_{uv} = E_{\text{dir}}(d_{uv}) + E_{\text{slot}}(s_{uv}) + E_{\text{role}}(r_{uv}), \quad (1)$$

$$\widetilde{h}_v^{(\ell+1)} = \text{MLP}_\ell\left((1+\epsilon_\ell)\, h_v^{(\ell)} + m_v^{(\ell)}\right),$$
$$m_v^{(\ell)} = \sum_{u \in \mathcal{N}(v)} \text{ReLU}\left(h_u^{(\ell)} + W_\ell\, a_{uv}\right), \quad (2)$$

with the exact operation order fixed by the frozen implementation. The four arms differ only in their input (AST-DAG, pure-EML-DAG, frequent-motif DAG, motif-AST). Every compressed graph is required to reconstruct its source expression exactly.

4.2 Task composition and controls

For symmetric equivalence classification, the shared endpoint encoders produce graph embeddings g_a, g_b, combined as

$$p(a, b) = \left[\, g_a + g_b,\ |g_a - g_b|,\ g_a \odot g_b \,\right], \quad (3)$$

symmetric in a, b by construction. The transparent floor uses 19 predeclared surface counts that every learned arm must beat: counts of operators, primitives, and variables, plus size and depth. Vocabulary embeddings count toward the matched parameter budget, since pure-EML label vocabularies differ per channel and shift the totals.

4.3 Compute matching and training

Because pure-EML graphs are much larger than AST or macro graphs for the same expression, equal parameters are *not* equal compute. We therefore report, jointly with every quality number, the full compute envelope (parameters, FLOPs, wall time, peak memory, GPU-hours, graph/token size, batch and step budgets, early-stopping state, precision, and deterministic settings). Hidden width and the virtual-node flag are fixed by a measured parameter/FLOP preflight on the production vocabularies; until that preflight runs, the grid configuration records the choice as pending rather than guessing it (Table 3, Appendix A). We use three seeds and publish every seed row; the mean never replaces the raw rows. The default profile is 2×H100 80 GB.

5 Bounded tasks and verification

We evaluate representation quality through bounded, verifier-gated tasks rather than a single aggregate score. Every task has frozen identifiers, declared budgets, retained failure states, and explicit denominators; unavailable cells are reported as missing, not encoded as zero.

5.1 Equivalence

The production equivalence specification fixes 50,000/5,000/5,000/5,000 accepted pair records (train / validation / test_iid / test_ood). Each accepted pair is a concrete, replayable rewrite trajectory: rule identifier, direction, occurrence path, bindings, assumptions, source and successor signatures, and separate verifier evidence. Source-, e-class-, and trace-relative grouping prevents leakage across partitions, and rejected, invalid, unsupported, and timeout candidates are kept in typed ledgers. The build aligns every endpoint across the four graph channels and two controls of §3; IID, OOD-stress, depth-OOD, and family-OOD views never alter the base split totals.

The pipeline has run end-to-end on a 10,000-row pilot corpus (regenerated byte-identically twice) using the real rule engine and replay verifier, accepting 9,039 records in 34 CPU minutes (6,867 replay-verified positives, 2,172 exact-rationally refuted negatives) and retaining 2,824 rejections and 3,133 typed failures. The run is recorded as complete but short of the production pair targets, and the shortfall is reported rather than resampled away (Table 5). Its `test_ood` split holds a single accepted negative, because the stress family offers few same-family size-matched neighbors; we retain and report the imbalance.

5.2 Further bounded tracks (future work)

The remaining tracks are frozen to the same standard and scheduled as future work; we report no numbers from them here. Rewrite-step prediction fixes one immutable 6-arm × 3-seed design over the registry-derived legal-action mask, reporting demonstration-action and exact-successor top-k separately with complete attempted/valid/invalid/unsupported/timeout denominators; its first GPU run is partial (13 of 18 logical cells, §6.5) and decides no gate. The proof benchmark (256 frozen problems) and simplification benchmark (1,000 frozen identifiers) count success only for a replayed, verifier-confirmed trace reaching the exact target signature, with every unsuccessful row charged at least its expanded-node budget. The symbolic-regression benchmark (256 frozen synthetic tasks plus a restricted 32-task Feynman-style subset (Udrescu and Tegmark, 2020)) takes verifier-confirmed recovery as its sole criterion; its manifests reproduced identically across three independent CPU builds. A capstone Pareto/Gate-G11 synthesis is specified to compose the track outcomes into a single frozen verdict without retraining; it has not run. These specifications ship with the apparatus.

6 Results at fixed scale

Table 1 collects every headline structural number; each row names its committed provenance document. All values come from real CPU production runs recorded in those documents; none is estimated. We organize by outcome rather than by a single leaderboard.

6.1 Raw expansion at production scale

Compiling all 250,000 expressions with the pinned official-v4 compiler succeeded on every row and produced a median tree expansion factor $\alpha = 40.6602$ over the source AST (mean 952.1371; full distribution in Appendix B).[2] The mean/median gap is a heavy right tail dominated by the trig/hyperbolic family (median 270.46), not a typical 952-fold expansion. Against the six preregistered per-family thresholds $1 + \ln K/\ln(4L)$ (1.2914–1.5000) and a whole-grammar counting break-even of $\approx$1.56, the strict pass count is 0/250,000: no expression in any family is close to the regime where a reduced operator vocabulary pays for its node count (Table 2). These thresholds are descriptive combinatorial references, not performance laws. The informative content of the zero is therefore the size of the miss, one to two orders of magnitude in every family, not the pass count itself.

6.2 Lossless sharing and the remaining gap

Lossless hash-consed sharing compresses expanded EML trees by a mean factor of 39.3750, yet the shared EML DAG remains larger than the *uncompressed* AST tree for every one of the 250,000 expressions; the best remaining ratio is exactly 8/7.[3] The most compressible expression shrinks by 28,087,277/180 and still ends 540/13 times larger than its AST (mean EML-DAG/AST-tree 8.3344, EML-DAG/AST-DAG 10.4750). "Compresses well" and "becomes structurally competitive" are different claims, and only the first is true here.

6.3 Verified rewriting adds conditional savings

On a frozen, stratified 30,000-expression selection, we run bounded equality saturation (Tate et al., 2009; Willsey et al., 2021) with complete application provenance and independent candidate verification. It improves 4,349/18,210 costed rows (23.882%) under branch-insensitive real rewrites (`safe_real`) and 5,026/18,210 (27.600%) with guarded positive-real rules (`positive_real_formal`), at mean relative savings of 2.361% and 2.849% over costed rows and a largest single reduction of 302 exact EML-DAG nodes.[4] Coverage is the binding caveat: 11,566 rows per mode (38.553%) contain operators outside the closed rewrite vocabulary (including *all* trig/hyperbolic and mixed-elementary rows), so these results hold on 60.700% cost coverage and are conditional on it. Each mode retains 224 validation failures (0.747%) and zero timeouts; the source form is always a costed safety anchor, so improvement rates cannot be inflated by dropped rows.

6.4 Learned components do not yet beat transparent baselines

Both Goal 5 comparisons were published in the frozen null-before-positive order.[5] First, a learned motif vocabulary did not beat the equal-budget frequent-motif vocabulary on `test_iid` dictionary-inclusive MDL: 324,485,346 vs. 317,678,264 bits over 25,000 expressions (it did beat the uncompressed macro baseline and the median of 30 fixed random vocabularies). Second, a neural cost ranker did not beat every structural heuristic under the frozen comparison order: 8,349/18,050 exact-best rankings vs. 8,796/18,050 for the estimated-EML-tree-cost heuristic (and 8,661/18,050 for the AST-DAG-cost heuristic). Counting is conservative: of the 18,050 groups, 10,752 were evaluable, and every unevaluable group scores as a miss for every ranker. On these tasks the transparent

[2] Supplementary bundle: Goal 2 summary, "Result" and "Expansion distribution".

[3] Supplementary bundle: Goal 3 summary, "Structural verdict".

[4] Supplementary bundle: Goal 4 summary, "Production result".

[5] Supplementary bundle: Goal 5 summary, "Learned and neural comparisons".

Quantity	Value	Provenance
Rows processed / exact-count successes	250,000 / 250,000	Goal 2
Median tree α (EML tree / AST tree)	**40.6602**	Goal 2
Mean tree α	952.1371 (σ = 23,155.56)	Goal 2
p90 / p99 / max α	385.07 / 10,448.60 / 6,481,679.31	Goal 2
Preregistered per-family thresholds 1.2914–1.5000: strict passes	**0 / 250,000**	Goal 2
Worst family median α (trig/hyperbolic, 40,000 rows)	270.4602	Goal 2
Mean EML self-compression, tree→DAG	39.3750×	Goal 3
Mean EML DAG / AST tree	**8.3344**	Goal 3
Mean EML DAG / AST DAG	10.4750	Goal 3
Expressions with EML DAG $\leq$ AST tree	**0 / 250,000** (best exactly 8/7)	Goal 3
Largest single self-compression (remaining ratio)	28,087,277/180 (540/13)	Goal 3
E-graph cost coverage	18,210 / 30,000 (60.700%)	Goal 4
Improved / costed, `safe_real`	4,349 / 18,210 (23.882%)	Goal 4
Improved / costed, `positive_real_formal`	5,026 / 18,210 (27.600%)	Goal 4
Mean relative savings over costed rows (incl. unchanged)	2.361% / 2.849%	Goal 4
Unsupported-operator / validation-failure / timeout rows per mode	11,566 / 224 / 0	Goal 4
Null: learned motif vocab. vs. equal-budget frequent (MDL bits, test_iid)	324,485,346 vs. **317,678,264**	Goal 5
Null: neural ranker vs. structural heuristic (exact-best of 18,050)	8,349 vs. **8,796**	Goal 5

Table 1: All measured structural results at production scale. Bold marks the winning side or the decisive count. Each row's provenance is the corresponding per-goal summary in the supplementary bundle; production artifacts are stored off-repository under recorded SHA-256 digests.

Family	Rows	Median α	Thr.
algebraic core	70,000	25.5778	1.3010
exp/log	40,000	32.7872	1.3382
powers/div/rationals	40,000	44.1579	1.2914
OOD stress	25,000	45.7377	1.3382
mixed elementary	35,000	133.9289	1.4292
trig/hyperbolic	40,000	270.4602	1.5000

Table 2: Per-family median expansion α against the preregistered strict threshold. Every family misses by one to two orders of magnitude; the strict pass count is 0 in each (0/250,000 overall).

structural baselines remain unbeaten at equal budget—the standard any learned component must now clear.

6.5 Learning tracks: first verdicts

The frozen apparatus has now produced its first verdicts: every number in this subsection binds to the committed, digest-verified GPU run ledger. Of the five tracks, two are measured (equivalence and the value head, below); rewrite-step prediction is partial, at 13 of 18 logical cells (5 invalid and retained; retrieval grid 15/15); symbolic regression and the Pareto/Gate-G11 synthesis were not run and are scoped as future work (§5.2). The two measured tracks yield one viability verdict and one methodological verdict.

The equivalence track is a viability result, and we bound it carefully. Trained purely on single-operator EML-DAG source cells (three seeds, no AST comparison arm), two seeds reach validation BCE losses of 0.5267 and 0.5387 while the third diverges to 2.3914. The operative chance floor is not the balanced $\ln 2 \approx 0.693$: accepted pairs are label-imbalanced by construction and never resampled (the pilot build's validation split is 73.6% positive; Table 5), and a constant majority-rate predictor already attains ≈0.577 BCE on that composition. Against the imbalance-adjusted floor, the converged seeds clear it by 0.04–0.05 nats: learning, though modest. The frozen cell schema retains the loss alone (no accuracy or AUC), so we can state *that* the cells learn, not how well they classify; we record the gap rather than backfill it. That a 40×-expanded,

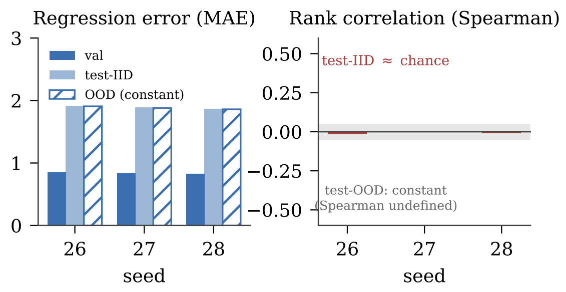

Figure 1: The value-head diagnostics: the rank and OOD probes expose what error metrics alone would miss. Targets are remaining witness steps, bounded above by four. **Left:** regression error (validation MAE 0.847/0.838/0.830, test-IID MAE 1.92/1.89/1.87 across seeds 26/27/28); the hatched bar is the model's own *constant* OOD predictor at MAE 1.91/1.88/1.86—indistinguishable from the trained test-IID error. **Right:** the same model ranks at chance (test-IID Spearman $-0.014/0.002/-0.009$, ≈ 0, grey band), and the constant OOD predictor's Spearman is undefined. Error alone would pass it; the rank and OOD probes fail it.

maximally regular graph admits any learning shows homogeneity is not by itself an obstacle to a graph model. It is not evidence that EML helps: the discriminative six-arm EML-vs-AST grid was not run, and the encoder width and virtual-node choice remain pending the measured preflight. The viability question is answered in this weak sense; the comparison stays open.

The value-head track is where the preregistered diagnostics prove their value (Figure 1). The head predicts *remaining witness steps*—for an ordered (current, goal) pair, how many rewrite steps remain on a stored, replayable witness—so targets in this run are bounded above by the maximum witness length of four. Under a reduced budget (1,500 optimizer steps, three seeds, 452,820 parameters, family `exp_log` held out), the goal-conditioned head reaches a validation MAE of 0.830–0.847 and a test-IID MAE of 1.87–1.92, numbers a proxy-metric report would pass. The diagnostics fail it: test-IID Spearman is -0.014, 0.002, and -0.009 across seeds, indistinguishable from zero, and the out-of-distribution predictor is *constant* (Spearman undefined), with the constant attaining MAE 1.86–1.91, indistinguishable from the trained head's own test-IID error. On this task, error magnitude cannot even separate the trained head from a constant one; the rank and OOD probes can. The run is recorded as a null, a collapsed value head: a guidance signal that error metrics alone would have passed. We report it as the methodological finding it is, not a working proof-search heuristic.

7 Discussion

The value-head diagnostics are the paper's most transferable export. In neural-guided proof and symbolic-regression search, a learned value or cost head is routinely selected and reported by a proxy regression loss (Polu and Sutskever, 2020; Lample et al., 2022). Ours shows unremarkable regression error yet ranks its own held-out states at chance and degenerates to a constant out of distribution. The constant is no strawman: its error matches the trained head's own test-IID error. Judged by the usual proxy metric it would have shipped as a working heuristic. It was caught only because rank and out-of-distribution diagnostics were preregistered alongside the error metric; both ship with the released apparatus. The three findings compose into one account: the structural cost is now exactly mapped, the representation is learnable on binary equivalence despite it, and usable guidance is the open frontier. Under bounded compute, regularity has bought learnability before guidance; closing that gap is where single-operator work should look next, and the released apparatus is frozen to test exactly that.

8 Limitations, integrity, and reproducibility

One scale. Everything is measured at one fixed scale, the 250,000-expression v1 corpus; we fit no scaling law and extrapolate to no other corpus, budget, or grammar. The expressions are synthetic and generated, and the downstream benchmarks are bounded, which limits external validity. With three seeds, learning effects are descriptive; we make no statistical-significance claims.

Bounded semantic audit. Numeric semantic verification covers a deterministic sample, not the whole corpus. Across the original Goal 2 audit (273 rows) and a widened pilot audit (2,988 rows), the combined evidence base is 3,261 rows with 6 retained mismatches ($\approx$0.1%).[6] The widened audit also hit one row whose numeric materialization deterministically hung (reproduced twice), a correctness finding that motivates a per-row timeout. We claim structural determinism corpus-wide but numeric verification only for the audited sample; no mismatch is relabeled or resampled away.

Conditional coverage and equal-compute. The e-graph results hold on 60.7% cost coverage, and the excluded families include trig/hyperbolic, the worst-expansion family, so the semantic-compression numbers are not corpus-wide. Because pure-EML expansion changes input size, equal parameters are not equal compute in the learning tracks, and we report both sides.

Weak learning margins. The equivalence verdict rests on two of three seeds clearing an imbalance-adjusted floor by 0.04–0.05 nats, on pilot-scale pairs; the frozen record retains only the loss, and we do not manufacture accuracy post hoc. The value-head verdict rests on one reduced-budget configuration; a larger budget could in principle train past the collapse, and we claim only what the measured run shows.

Grammar-v2 conformance (Gate G10) and external context. The grammar-v2 workstream adds opt-in `asin`/`acos`/`atan`/π/e compiler support with the v1 default untouched. Its preregistered bounded conformance audit (74 retained rows, all quotas met, no dropped denominators) records a fail: zero integrity, coverage, or v1-drift failures, but the eight preregistered `asin`/`acos` endpoint cells are nonfinite at 100-digit precision under the predeclared 10^{-60} tolerances, byte-identical across three executions.[7] Endpoint cases are required valid inputs under the preregistration, so the gate records the miss rather than lowering precision until they appear finite (no criteria revision is pre-authorized). The verdict bounds only the grammar-v2 surface, stays unresolved by design, and neither enables nor blocks any corpus or learning result. A frontier-LLM reference panel is implemented with full provenance capture but remains external context: proprietary data, nondeterminism, and model drift keep it out of every controlled gate.

Artifacts and regeneration. Production artifacts are stored off-repository; the repository commits their SHA-256 digests, per-goal manifests, exact CPU reproduction commands, and a single frozen lock file verified by a CPU-only CI. Generation reproduces on commodity hardware (the 10,000-row pilot regenerated byte-identically twice under a recorded corpus hash), and the 250,000-row stage refuses a machine below a twofold memory margin. Artifact authentication is staged: the structural (Goals 1–5) archive is held off-repository, so its per-claim digest index is committed ahead of final authentication, while the learning verdicts of §6.5 are authenticated directly against the committed GPU run ledger.

9 Conclusion

At one fixed 250,000-expression scale, this study delivers the single-operator representation's first exact cost profile, its first learning verdicts, and a frozen apparatus for the work ahead. The map is exact: median 40.66$\times$ tree expansion, 39.4$\times$ mean recovery from lossless sharing, and a further 2.4–2.8% from verified rewriting on 60.7% coverage, every number bound to committed provenance. The verdicts answer the viability question: despite the expansion, two of three equivalence seeds clear the imbalance-adjusted $\approx$0.58 chance floor, and the preregistered rank and out-of-distribution diagnostics pinpoint what remains to be learned, namely usable search guidance. That frontier is precisely what the released tracks are frozen to test.

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[6] Supplementary bundle: Goal 2 summary, "Semantic audit and retained issues".

[7] Supplementary bundle: Goal 10 summary, "Published verdict".

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A Preflight encoder parameter counts

Width	Virt. node	Params	Target
64	no	185,732	below
64	yes	210,692	within
96	no	343,044	within
96	yes	398,916	within

Table 3: Measured encoder parameter counts on the fixture vocabulary (Goal 6 preflight). Width-64 without a virtual node sits marginally under the $\approx$0.2M bound—recorded, not padded; production vocabularies will move these totals, which is why vocabulary embeddings count toward the match.

B Distribution of EML Tree Expansion Factors

Table 4 summarizes the distribution of the tree expansion factor α across all 250,000 expressions in the Goal 2 production run (a clean pinned commit, recorded in the committed summary; eight workers, 2,557.26 s wall time). The largest observed expansion occurred for an expression with 13 AST nodes, which compiled to 84,261,831 EML nodes, corresponding to $\alpha = 6{,}481{,}679.31$.

Statistic	**Tree α**
minimum	1.5000
p10	21.0338
p25	26.3800
median	40.6602
p75	75.2550
p90	385.0723
p95	1,187.8240
p99	10,448.5978
maximum	6,481,679.3077
mean	952.1371
std. dev.	23,155.5594

Table 4: Expansion factor α = (exact EML tree nodes)/(source AST nodes) over all 250,000 rows (Goal 2).

C Pilot pair build

Table 5 summarizes the pilot pair build of §5 per split. The run is recorded under a frozen, hash-identified configuration as complete but short: accepted counts fall short of the production targets of 50,000/5,000/5,000/5,000, and the shortfall is reported (the "Short" column) rather than adjusted by resampling or augmentation. Each positive pair carries a replayable verifier trace; negative pairs are validated by formal refutation at exact rational probe points. Converted to the unified graph-channel representation, the build yields 18,078 encoded endpoints per channel—two per accepted pair.

Split	**Acc. (pos+neg)**	**Rej.**	**Fail**	**Short**
train	6,281 (4,568+1,713)	1,786	2,432	43,719
validation	893 (657+236)	263	343	4,107
test_iid	864 (642+222)	276	358	4,136
test_ood	1,001 (1,000+1)	499	0	3,999
total	9,039 (6,867+2,172)	2,824	3,133	—

Table 5: Pilot pair build (Goal 6 pipeline). Accepted = positives + negatives; rejected and failed rows are retained with typed reasons; "Short" is the shortfall against the production split targets. The lone `test_ood` negative reflects few same-family size-matched neighbors in the stress family and is reported, not patched.